

Kaiwen Wu

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Research

I am interested in both theory and applications of machine learning. In my PhD, I have been working on various areas in probabilistic machine learning including Gaussian processes, variational inference, Bayesian optimization, and active learning. Before starting my PhD, I worked on generative modeling, convex optimization, and adversarial robustness of deep learning models.

Education

University of Pennsylvania	Philadelphia, PA
Doctor of Philosophy in Computer and Information Science	2021 – Present
Advisor: Jacob R. Gardner	

University of Waterloo	Waterloo, ON
Master of Mathematics in Computer Science (Thesis Option)	2018 – 2020
Advisor: Yaoliang Yu	

Nanjing University	Nanjing, China
Bachelor of Science in Computer Science	2014 – 2018

Work Experience

Meta	New York, NY
Research Scientist Intern	Jun 2025 – Oct 2025
• Worked on Bayesian optimization for automated machine learning.	

Meta	Redmond, WA
Research Scientist Intern	Aug 2024 – Dec 2024
• Worked on experimental designs for VR/AR hardware using active learning.	

Borealis AI	Waterloo, ON
Research Intern	Jan 2019 – May 2019
• Worked on robust mean estimation on noisy data using generative adversarial networks.	
• Published a paper at AISTATS 2020.	

Open Source Contributions

Ax	https://github.com/facebook/ax
An adaptive experimentation platform written in Python	
• Implemented new features supporting experiments in hierarchical search spaces.	

BoTorch	https://github.com/pytorch/botorch
A Python package for Bayesian optimization in PyTorch	
• Implemented an elliptical slice sampling algorithm for truncated normal distributions.	

GPyTorch	https://github.com/cornellius-gp/gpytorch
A Python package for Gaussian processes in PyTorch	
• Implemented stochastic variational Gaussian processes. Implemented custom autograd functions that yield 4.45x speed-up for large batch stochastic training on A100 GPUs.	
• Maintenance and review pull requests.	

Publications

See also at my Google Scholar .

Refereed Conference Proceedings

- [C1] Jonathan Wenger, **Kaiwen Wu**, Philipp Hennig, Jacob R. Gardner, Geoff Pleiss, and John P. Cunningham
Computation-Aware Gaussian Processes: Model Selection And Linear-Time Inference
Advances in Neural Information Processing Systems (NeurIPS). 2024.
- [C2] **Kaiwen Wu** and Jacob R. Gardner
Understanding Stochastic Natural Gradient Variational Inference
Proceedings of the 41st International Conference on Machine Learning (ICML). 2024.
- [C3] **Kaiwen Wu**, Jonathan Wenger, Haydn T Jones, Geoff Pleiss, and Jacob Gardner
Large-Scale Gaussian Processes via Alternating Projection
Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS). 2024.
- [C4] Kyurae Kim, Jisu Oh, **Kaiwen Wu**, Yian Ma, and Jacob R. Gardner
On the Convergence of Black-Box Variational Inference
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- [C5] Kyurae Kim, **Kaiwen Wu**, Jisu Oh, and Jacob R. Gardner
Practical and Matching Gradient Variance Bounds for Black-Box Variational Bayesian Inference
Proceedings of the 40th International Conference on Machine Learning (ICML). 2023.
- [C6] Natalie Maus, **Kaiwen Wu**, David Eriksson, and Jacob Gardner
Discovering Many Diverse Solutions with Bayesian Optimization
Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS). 2023.
- [C7] **Kaiwen Wu**, Kyurae Kim, Roman Garnett, and Jacob R. Gardner
The Behavior and Convergence of Local Bayesian Optimization
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- [C8] Xinran Zhu, **Kaiwen Wu**, Natalie Maus, Jacob Gardner, and David Bindel
Variational Gaussian Processes with Decoupled Conditionals
Advances in Neural Information Processing Systems (NeurIPS). 2023.
- [C9] Quan Nguyen, **Kaiwen Wu**, Jacob Gardner, and Roman Garnett
Local Bayesian optimization via maximizing probability of descent
Advances in Neural Information Processing Systems (NeurIPS). 2022.
- [C10] **Kaiwen Wu**, Gavin Weiguang Ding, Ruitong Huang, and Yaoliang Yu
On Minimax Optimality of GANs for Robust Mean Estimation
Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS). 2020.
- [C11] **Kaiwen Wu**, Allen Wang, and Yaoliang Yu
Stronger and Faster Wasserstein Adversarial Attacks
Proceedings of the 37th International Conference on Machine Learning (ICML). 2020.

Lightly Reviewed Workshop Papers

- [W1] **Kaiwen Wu** and Jacob R. Gardner
A Fast, Robust Elliptical Slice Sampling Method for Linearly Truncated Multivariate Normal Distributions
NeurIPS Workshop on Bayesian Decision-making and Uncertainty. 2024.
- [W2] Guojun Zhang, **Kaiwen Wu**, Pascal Poupart, and Yaoliang Yu
Newton-type Methods for Minimax Optimization
ICML Workshop on Beyond First-order Methods in Machine Learning Systems. 2021.

Preprints

- [P1] **Kaiwen Wu**, Craig Sanders, Benjamin Letham, and Phillip Guan
Mixed Likelihood Variational Gaussian Processes
arXiv preprint arXiv:2503.04138. 2025.

Teaching Experience

University of Pennsylvania

- Teaching Assistant, CIS 520 Machine Learning Fall 2023
- Teaching Assistant, ESE 204 Decision Models Fall 2022

University of Waterloo

- Teaching Assistant, CS 370 Numerical Computation Spring 2020
- Teaching Assistant, CS 486/686 Introduction to Artificial Intelligence Spring 2019
- Teaching Assistant, CS 480/680 Introduction to Machine Learning Fall 2018

Honors & Awards

- Top Reviewer, Conference on Neural Information Processing Systems 2024
- Graduate Fellowship, University of Pennsylvania 2021
- Vector Scholarship in Artificial Intelligence, Vector Institute 2018
- Entrance Scholarship, University of Waterloo 2018
- Jimin Liu Scholarship, Nanjing University 2017

Academic Service

Conference Reviews

- Program Committee Member, AAAI Conference on Artificial Intelligence (2021, 2026)
- Reviewer, Conference on Neural Information Processing Systems (2023 – 2025)
- Reviewer, International Conference on Artificial Intelligence and Statistics (2021, 2024)
- Reviewer, International Conference on Learning Representations (2024 – 2025)
- Reviewer, International Conference on Machine Learning (2023 – 2025)

Journal Reviews

- Reviewer, Transactions on Machine Learning Research (2025)
- Reviewer, Journal of Machine Learning Research (2025)